
Statistical Inference

B. Statistical Data Science 2nd Year Indian Statistical Institute

Mid-semester Examination

10 October 2025

11 am – 1 pm

Name:

Roll Number:

Instructions

- Answer as much as you can. The maximum you can score is 30.
 - You can use any result obtained in the class. Clearly state the results that you use.
 - Clearly state all the assumptions that you make. Failure to do so will lead to deduction of marks.
 - Keep your answers to the point. Any unnecessary or irrelevant writing will be penalized.
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Question 1.

Total marks: 12

The following is a snapshot of the yearly number of accidents at a coal mine in the UK.

Year	1851	1852	1853	1854	1855	...	1958	1959	1960	1961	1962
No. of accidents	4	5	4	1	0	...	0	0	1	0	1

We want to statistically model this data.

- (a) Formulate a suitable statistical model for the data. Give justification for the choice of your model. Specify the model parameter and the parameter space. [2 marks]
- (b) Identify a minimal sufficient statistic for the model parameter. [2 marks]
- (c) Find the method of moments estimator and the maximum likelihood estimator of the parameter. [2+2 marks]
- (d) Find the maximum likelihood estimator of the probability of at least one accident in a particular year. [4 marks]

Question 2.

Total marks: 14

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} f_\theta$, $\theta \in \mathbb{R}$, where

$$f_\theta(x) = \begin{cases} e^{-(x-\theta)} & \text{if } x \geq \theta, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Show that $X_{(1)} = \min\{X_1, \dots, X_n\}$ is a minimal sufficient statistic for θ . Verify whether $X_{(1)}$ is complete. [2+4 marks]
- (b) Find the method of moments estimator of θ . [3 marks]

- (c) Find the maximum likelihood estimator of θ . [3 marks]
- (d) Among the two estimators above, which one gives a better summarization? Justify. [2 marks]

Question 3.

Total marks: 10

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}(-\theta, \theta)$, $\theta > 0$.

- (a) Find the maximum likelihood estimator of θ . [4 marks]
- (b) Find the method of moments estimator of θ . [4 marks]
- (c) Among these two estimators, which one would you prefer and why? [2 marks]